

Homework 3 — Code Profiling Report

BMI 500 Introduction to Ethical Data Science and Informatics

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Coarse-grain Timing

We document the coarse-grain timing in Table 1. We find that the time, CPU, and memory usage monotonically increase as the dataset gets larger. In particular, the CPU usage is within the range of 93%-100%, this means that The `--cpus-per-task=1` request is correctly sized. Nevertheless, we find that the peak RSS is simply 1.2GB, which is far less than the memory we requested. Therefore, we should reduce the memory call to 4GB for future tests.

Table 1: Coarse-grain timing in seconds.

Datasets	pbmc3k	pbmc6k	pbmc10k
Wall clock time (s)	27.22	36.25	74.30
Percent of CPU (%)	93	98	100
Maximum resident set size (MB)	851.6	931.8	1,293.1

Instrumented Profiling

We document the instrumented profiling results in Table 2. We find that the number in “total running time” is close to the results in coarse-grain timing, which further verifies the reliability of our results. We find that “neighbors, UMAP, rank_gene_groups” are the three operations that take most computation time; the time consumption for “neighbors and rank_gene_groups” is also increasing with respect to the input data size. We will further elaborate on it in later sections.

Table 2: Instrumented profiling results in seconds.

Datasets	pbmc3k	pbmc6k	pbmc10k	share @10K
imports	3.790	2.748	2.459	3.4 %
scanpy settings	7.456	7.268	7.132	9.7 %
argument parsing	0.001	0.001	0.001	0.0 %
I/O	0.158	0.192	0.672	0.9 %

Datasets	pbmc3k	pbmc6k	pbmc10k	share @10K
filtering	0.059	0.083	0.424	0.6 %
normalization	0.058	0.086	0.465	0.6 %
highly variable genes	0.064	0.081	0.457	0.6 %
scale	0.042	0.076	0.172	0.2 %
PCA	0.351	0.572	1.076	1.5 %
neighbors	4.191	4.231	30.539	41.7 %
louvain clustering	0.143	0.386	0.781	1.1 %
UMAP	8.291	16.079	14.182	19.4 %
write h5ad	0.156	0.210	0.552	0.8 %
rank_genes_groups	1.758	3.539	14.325	19.6 %
TOTAL (instrumented)	26.516	35.551	73.237	100 %

Fine-grain Profiling

We visualize the c-profiling results in Figure1, Figure 2 and Figure 3 for each dataset. We find that within the rank-gene-groups function, the most time-consumption step is the rank computation. In specific, Scanpy's `_ranks()` calls `pandas.core.algorithms.rank`, which dominates the profile in every dataset.

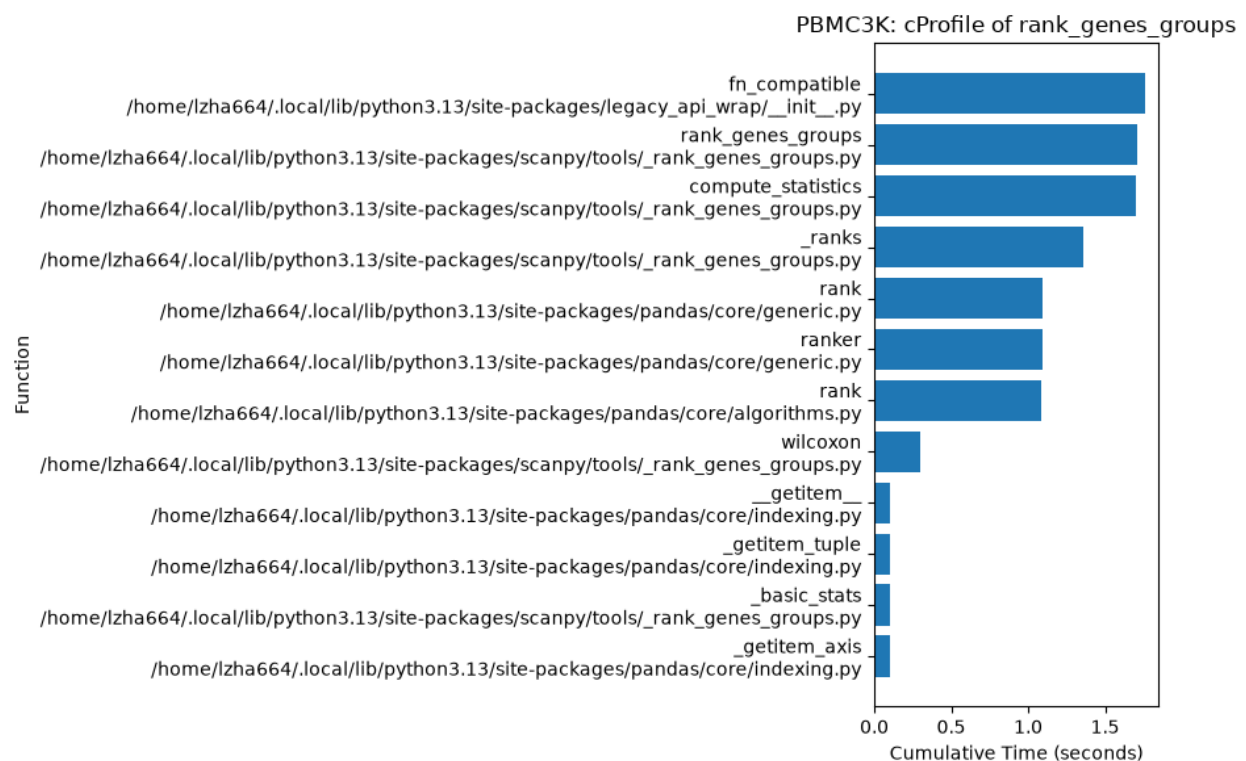


Figure 1: C-Profile for PBMC3K for rank-gene-group.

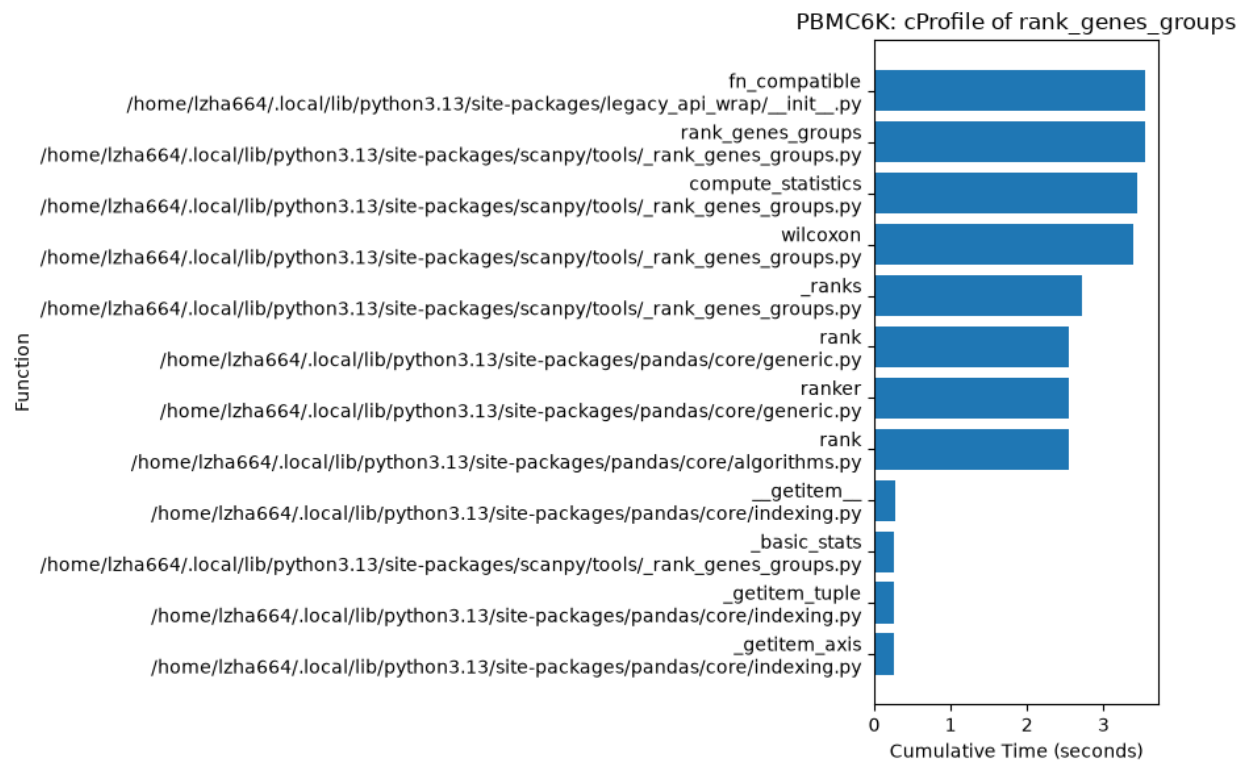


Figure 2: C-Profile for PBMC6K for rank-gene-group.

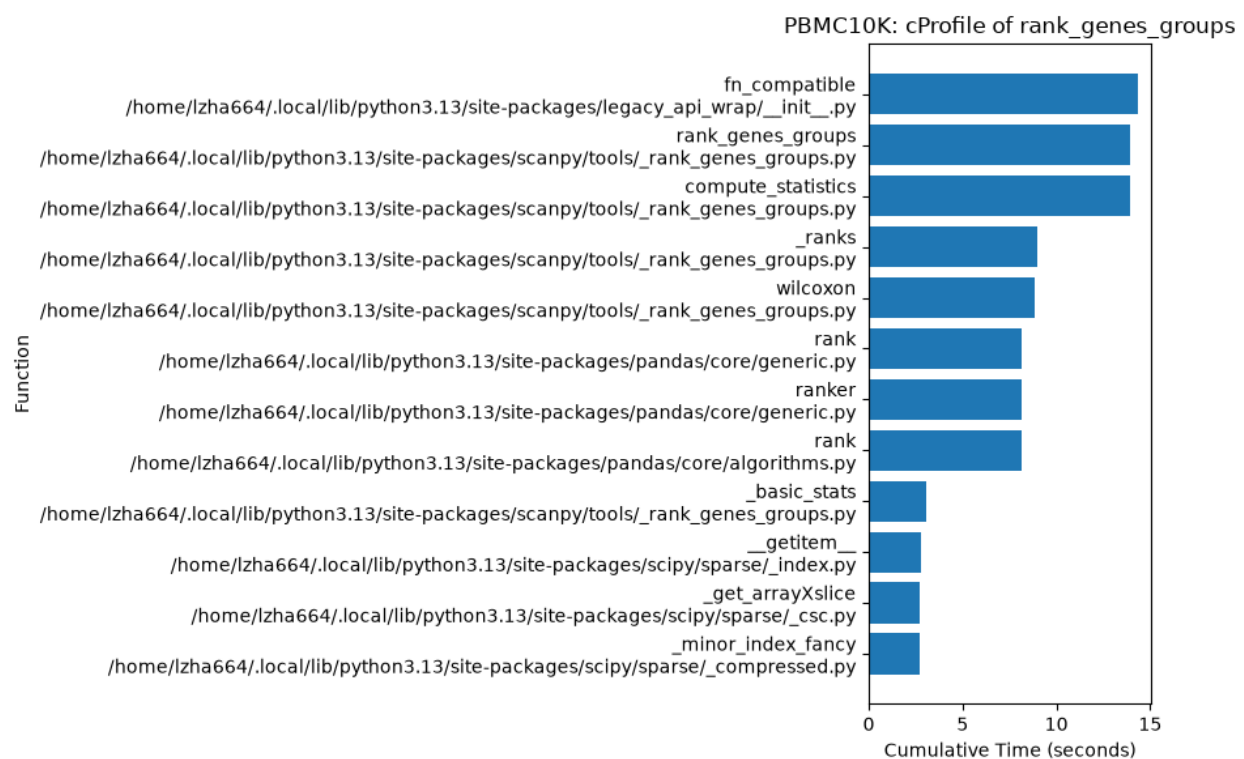


Figure 3: C-Profile for PBMC10K for rank-gene-group.

Runtime growth vs. dataset size

We visualize the time consumption with respect to different steps in Figure 4 and Figure 5. We find that some code sections (in particular, “neighbors, UMAP, rank_gene_groups”) computing time increases as datasets get enlarged, while other code sections do not introduce significant computation overhead.

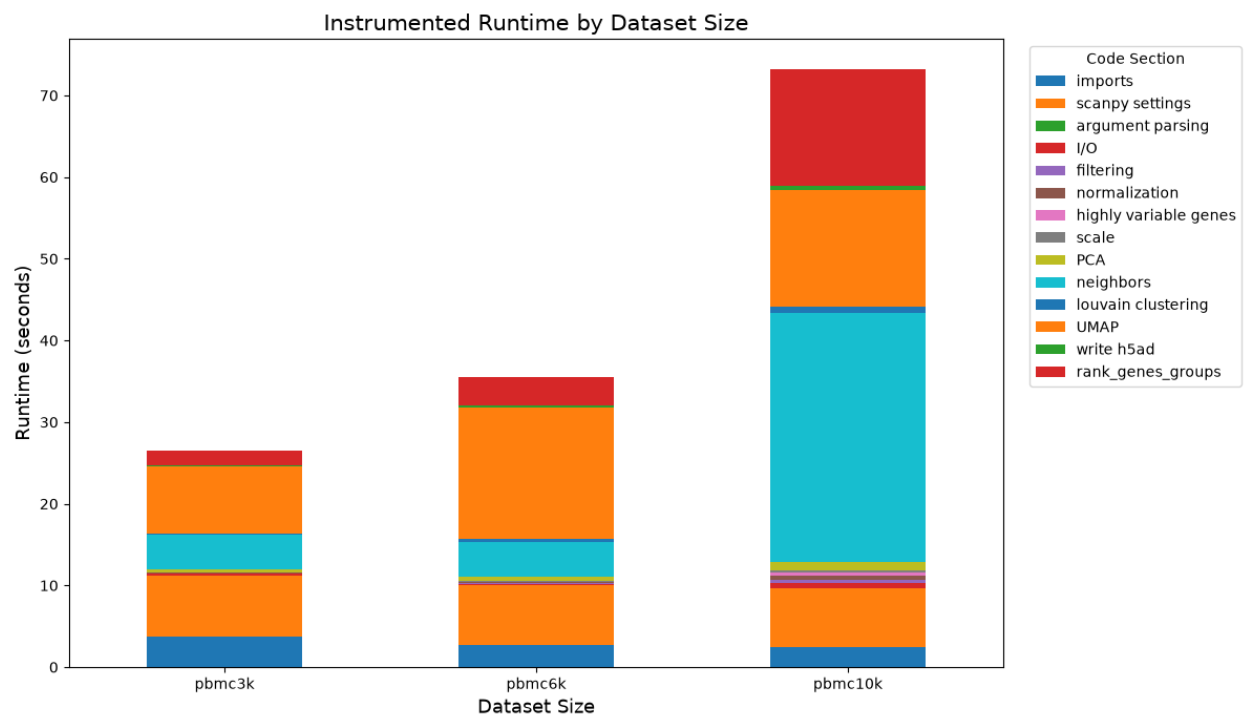


Figure 4: Stacked bar plot for computation time.

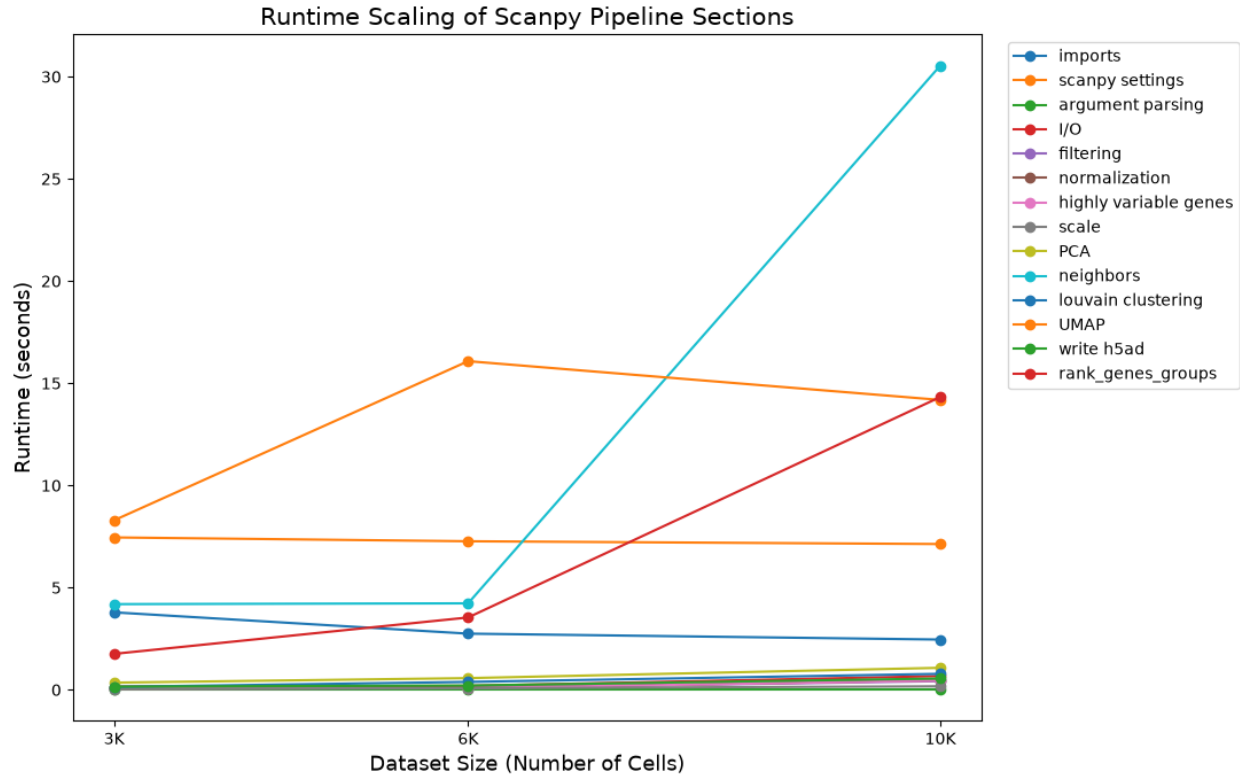


Figure 5: Line plot for computation time.

Bottleneck Analysis and Run-Time Prediction

We identified 2 bottlenecks in total. Specifically, we find that the “neighbors” is the dominant bottleneck at scale; “rank_genes_groups” is the fastest-growing section. In addition, “Umap” is the dominant bottleneck at small/medium scale, nevertheless, it does not seem to be the bottleneck when data size scales up.

Based on our observation, we conduct a least-squares linear regression of each section’s runtime on cell count, fitted over the three measured datasets and extrapolated to 20,000 cells. The result is shown in Table 3 below, and we estimate a run time of 138 seconds.

Table 3: Predicted runtime at PBMC20k by linear regression.

Section	Measured @10K (s)	Predicted @20K (s)	Share of prediction
neighbors	30.539	66.51	48.1 %
rank_genes_groups	14.325	31.74	22.9 %
UMAP	14.182	23.38	16.9 %
scanpy settings	7.132	6.66	4.8 %
PCA	1.076	2.10	1.5 %

Section	Measured @10K (s)	Predicted @20K (s)	Share of prediction
louvain clustering	0.781	1.69	1.2 %
I/O	0.672	1.38	1.0 %
write h5ad	0.552	1.10	0.8 %
normalization	0.465	1.02	0.7 %
highly variable genes	0.457	1.00	0.7 %
filtering	0.424	0.93	0.7 %
imports	2.459	0.49	0.4 %
scale	0.172	0.35	0.3 %
argument parsing	0.001	0.001	0.0 %
TOTAL	73.237	≈ 138.3	100 %

Appendix A. SLURM batch scripts (a, b)

Three identical scripts, differing only in `--job-name` and `--data-set`. Each requests 1 CPU, 16 GB and 5 minutes, and wraps the run in `/usr/bin/time -v` for coarse-grain metrics.

```
#!/bin/bash
#SBATCH --job-name=pbmc3k
#SBATCH --output=pbmc3k_%j.out
#SBATCH --error=pbmc3k_%j.err
#SBATCH --cpus-per-task=1
#SBATCH --mem=16G
#SBATCH --time=0:05:00

set -eo pipefail

cd "$SLURM_SUBMIT_DIR"

source ~/.bashrc
conda activate bmi500

echo "==== Job information ====="
echo "Dataset: pbmc3k"
echo "Node: $(hostname)"
echo "Start time: $(date)"
echo "Working directory: $(pwd)"
echo "===== "

/usr/bin/time -v python scanpy_pbmc.py \
    --data-dir data \
    --data-set pbmc3k \
```

```
--num-threads 1
```

```
echo "End time: $(date)"
```

submit_pbmc6k.sb and submit_pbmc10k.sb are identical, with pbmc3k replaced by pbmc6k and pbmc10k respectively, in the --job-name, --output, --error, echo "Dataset: ..." and --data-set fields.

```
#!/bin/bash
#SBATCH --job-name=pbmc6k
#SBATCH --output=pbmc6k_%j.out
#SBATCH --error=pbmc6k_%j.err
#SBATCH --cpus-per-task=1
#SBATCH --mem=16G
#SBATCH --time=0:05:00
```

```
set -eo pipefail
```

```
cd "$SLURM_SUBMIT_DIR"
```

```
source ~/.bashrc
conda activate bmi500
```

```
echo "==== Job information ====="
echo "Dataset: pbmc3k"
echo "Node: $(hostname)"
echo "Start time: $(date)"
echo "Working directory: $(pwd)"
echo "=====
```

```
/usr/bin/time -v python scanpy_pbmc.py \
  --data-dir data \
  --data-set pbmc6k \
  --num-threads 1
```

```
echo "End time: $(date)"
```

```
#!/bin/bash
#SBATCH --job-name=pbmc10k
#SBATCH --output=pbmc10k_%j.out
#SBATCH --error=pbmc10k_%j.err
#SBATCH --cpus-per-task=1
#SBATCH --mem=16G
#SBATCH --time=0:05:00
```

```
set -eo pipefail

cd "$SLURM_SUBMIT_DIR"

source ~/.bashrc
conda activate bmi500

echo "==== Job information ====="
echo "Dataset: pbmc10k"
echo "Node: $(hostname)"
echo "Start time: $(date)"
echo "Working directory: $(pwd)"
echo "===== "

/usr/bin/time -v python scanpy_pbmc.py \
    --data-dir data \
    --data-set pbmc10k \
    --num-threads 1

echo "End time: $(date)"
```

Appendix B. Console output (c)

pbmc3k (job 1290038)

stdout — pbmc3k_1290038.out

```
==== Job information =====
Dataset: pbmc3k
Node: bignode2.priv.bmi.emory.edu
Start time: Wed Sep 16 08:30:51 EDT 2026
Working directory: /home/lzha664/bmi500_hw2
=====
[timing] imports: 3.790 s
scanpy==1.10.2 anndata==0.12.19 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pand
as==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph==0.11.6 louvain==0.
8.2 pynndescent==0.6.0
using 1 threads
[timing] scanpy settings: 7.456 s
[timing] argument parsing: 0.001 s
[timing] I/O: 0.158 s
[timing] filtering: 0.059 s
[timing] normalization: 0.058 s
[timing] highly variable genes: 0.064 s
[timing] scale: 0.042 s
[timing] PCA: 0.351 s
[timing] neighbors: 4.191 s
```

```
[timing] louvain clustering: 0.143 s
[timing] UMAP: 8.291 s
[timing] write h5ad: 0.156 s
[timing] rank_genes_groups: 1.758 s
[cProfile] saved to pbmc3k_rank_genes_groups.prof
```

```
=== elapsed time per section ===
 3.790 s imports
 7.456 s scanpy settings
 0.001 s argument parsing
 0.158 s I/O
 0.059 s filtering
 0.058 s normalization
 0.064 s highly variable genes
 0.042 s scale
 0.351 s PCA
 4.191 s neighbors
 0.143 s louvain clustering
 8.291 s UMAP
 0.156 s write h5ad
 1.758 s rank_genes_groups
26.516 s TOTAL (instrumented sections)
End time: Wed Sep 16 08:31:18 EDT 2026
```

Resource usage — pbmc3k_1290038.err (/usr/bin/time -v)

```
Command being timed: "python scanpy_pbmc.py --data-dir data --data-set pb
mc3k --num-threads 1"
User time (seconds): 24.32
System time (seconds): 1.17
Percent of CPU this job got: 93%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:27.22
Maximum resident set size (kbytes): 871992
Major (requiring I/O) page faults: 2456
Minor (reclaiming a frame) page faults: 162716
Exit status: 0
```

pbmc6k (job 1290039)

stdout — pbmc6k_1290039.out

```
===== Job information =====
Dataset: pbmc6k
Node: bignode2.priv.bmi.emory.edu
Start time: Wed Sep 16 08:30:59 EDT 2026
Working directory: /home/lzha664/bmi500_hw2
=====
[timing] imports: 2.748 s
scanpy==1.10.2 anndata==0.12.19 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pand
as==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph==0.11.6 louvain==0.
8.2 pyndescent==0.6.0
```

```
using 1 threads
[timing] scanpy settings: 7.268 s
[timing] argument parsing: 0.001 s
[timing] I/O: 0.192 s
[timing] filtering: 0.083 s
[timing] normalization: 0.086 s
[timing] highly variable genes: 0.081 s
[timing] scale: 0.076 s
[timing] PCA: 0.572 s
[timing] neighbors: 4.231 s
[timing] louvain clustering: 0.386 s
[timing] UMAP: 16.079 s
[timing] write h5ad: 0.210 s
[timing] rank_genes_groups: 3.539 s
[cProfile] saved to pbmc6k_rank_genes_groups.prof
```

```
=== elapsed time per section ===
 2.748 s imports
 7.268 s scanpy settings
 0.001 s argument parsing
 0.192 s I/O
 0.083 s filtering
 0.086 s normalization
 0.081 s highly variable genes
 0.076 s scale
 0.572 s PCA
 4.231 s neighbors
 0.386 s louvain clustering
16.079 s UMAP
 0.210 s write h5ad
 3.539 s rank_genes_groups
35.551 s TOTAL (instrumented sections)
End time: Wed Sep 16 08:31:35 EDT 2026
```

Resource usage — pbmc6k_1290039.err (/usr/bin/time -v)

```
Command being timed: "python scanpy_pbmc.py --data-dir data --data-set pb
mc6k --num-threads 1"
User time (seconds): 34.64
System time (seconds): 1.23
Percent of CPU this job got: 98%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:36.25
Maximum resident set size (kbytes): 954116
Major (requiring I/O) page faults: 13
Minor (reclaiming a frame) page faults: 203463
Exit status: 0
```

pbmc10k (job 1290040)

stdout — pbmc10k_1290040.out

```
===== Job information =====
Dataset: pbmc10k
Node: bignode2.priv.bmi.emory.edu
Start time: Wed Sep 16 08:31:03 EDT 2026
Working directory: /home/lzha664/bmi500_hw2
=====
[timing] imports: 2.459 s
scanpy==1.10.2 anndata==0.12.19 umap==0.5.12 numpy==1.26.4 scipy==1.17.1 pand
as==2.3.3 scikit-learn==1.9.1 statsmodels==0.15.0 igraph==0.11.6 louvain==0.
8.2 pynndescent==0.6.0
using 1 threads
[timing] scanpy settings: 7.132 s
[timing] argument parsing: 0.001 s
[timing] I/O: 0.672 s
[timing] filtering: 0.424 s
[timing] normalization: 0.465 s
[timing] highly variable genes: 0.457 s
[timing] scale: 0.172 s
[timing] PCA: 1.076 s
[timing] neighbors: 30.539 s
[timing] louvain clustering: 0.781 s
[timing] UMAP: 14.182 s
[timing] write h5ad: 0.552 s
[timing] rank_genes_groups: 14.325 s
[cProfile] saved to pbmc10k_rank_genes_groups.prof

=== elapsed time per section ===
    2.459 s imports
    7.132 s scanpy settings
    0.001 s argument parsing
    0.672 s I/O
    0.424 s filtering
    0.465 s normalization
    0.457 s highly variable genes
    0.172 s scale
    1.076 s PCA
   30.539 s neighbors
    0.781 s louvain clustering
   14.182 s UMAP
    0.552 s write h5ad
   14.325 s rank_genes_groups
   73.237 s TOTAL (instrumented sections)
End time: Wed Sep 16 08:32:17 EDT 2026
```

Resource usage — pbmc10k_1290040.err (/usr/bin/time -v)

```
Command being timed: "python scanpy_pbmc.py --data-dir data --data-set pb
mc10k --num-threads 1"
User time (seconds): 70.21
System time (seconds): 4.14
```

Percent of CPU this job got: 100%
Elapsed (wall clock) time (h:mm:ss or m:ss): 1:14.30
Maximum resident set size (kbytes): 1324180
Major (requiring I/O) page faults: 9
Minor (reclaiming a frame) page faults: 715339
Exit status: 0